

Algebra 1
Unit 11
Lesson 1 Homework

Name Answer Key
Date _____
Period _____

1. In the equation $y = a(b)^x$, a represents the ☐ initial amount.
☒ growth factor. and b represents ☒ initial amount
☐ growth factor
2. When b is greater than 1, the equation models ☒ exponential growth.
☐ exponential decay.
3. When b is between 0 and 1, the equation models ☐ exponential growth.
☒ exponential decay.

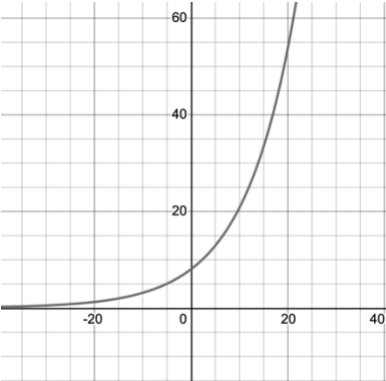
For questions 4-8, identify if each representation of a function is an example of exponential growth or exponential decay. Then justify your selection.

4.

x	y
-1	12
0	4
1	$\frac{4}{3}$
2	$\frac{4}{9}$
3	$\frac{4}{27}$

 represents exponential decay because...
multiply by factor of $\frac{1}{3}$

5. $y = 25(0.79)^x$ represents exponential decay because...
 b is 0.79 which is < 1

6.  represents exponential growth because...
the y -values increase

7. $y = 6(1 + 0.05)^x$ represents exponential growth because...
 $b > 1$

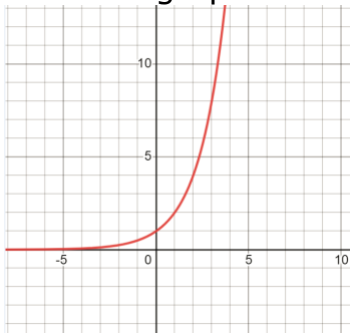
8.

x	y
2	9
3	13.5
4	20.25
5	30.375

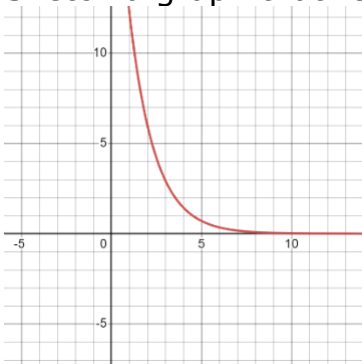
 represents exponential growth because...

increases by rate of 1.5

9. Sketch a graph that represents exponential growth.



10. Sketch a graph that represents exponential decay.



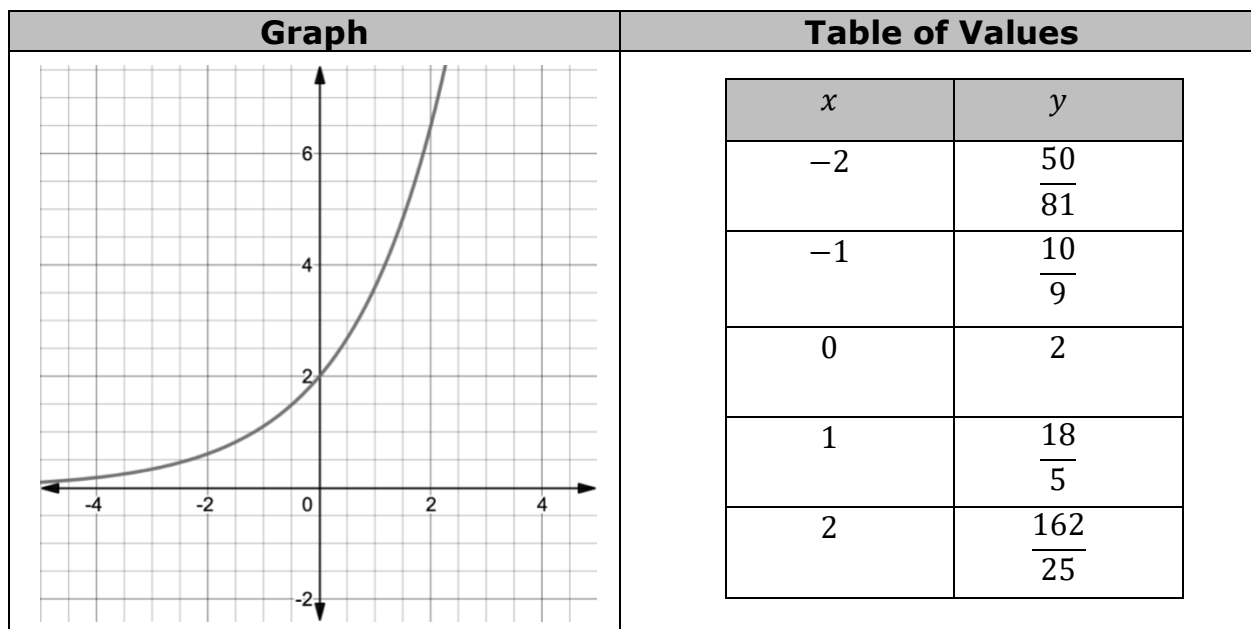
Algebra 1
Unit 11
Lesson 2 Homework

Name Answer Key
Date _____
Period _____

1. The graph of $y = \left(\frac{5}{6}\right)^x$ is ☐ increasing and is ☒ positive.
☒ decreasing ☐ negative.

2. The graph of $y = 2(1 + 0.25)^x$ is ☒ increasing and is ☒ positive.
☐ decreasing ☐ negative.

Use the graph and table of $y = 2\left(\frac{9}{5}\right)^x$ to determine key features.



Key Features			
3. Domain	All real numbers	4. Range	$y > 0$
5. y- intercept	(0,2)	6. Constant Percent Rate of Change	1.8 180%
7. Increasing or Decreasing	Increasing	8. Positive	$(-\infty, \infty)$
		Negative	n/a
9. Left End Behavior	As $x \rightarrow -\infty, y \rightarrow 0$	10. Right End Behavior	As $x \rightarrow \infty, y \rightarrow \infty$

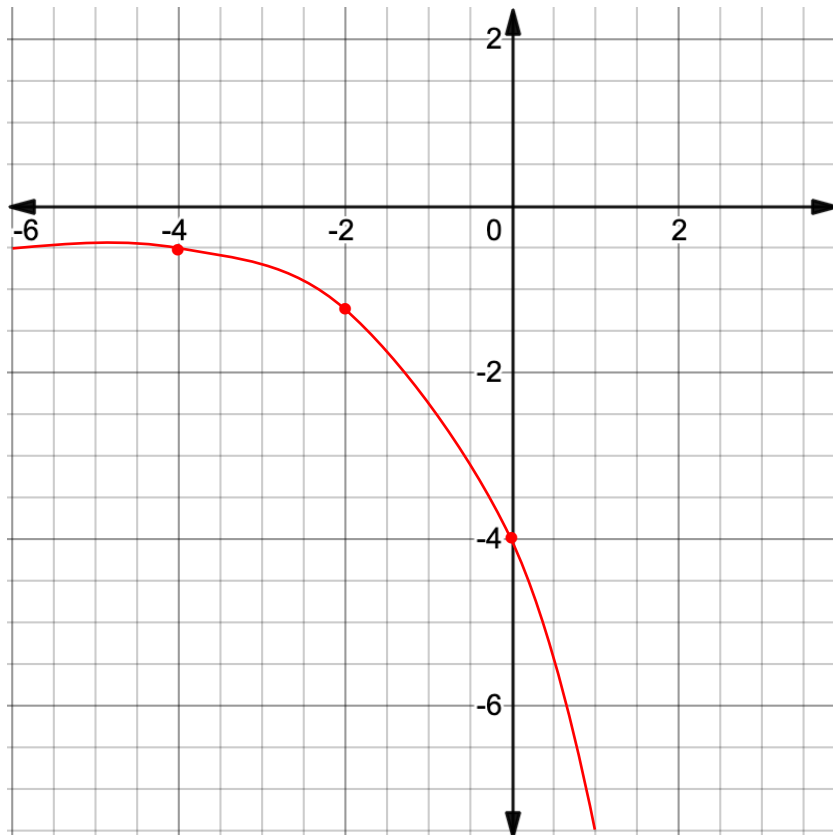
Algebra 1
Unit 11
Lesson 3 Homework

Name **Answer Key** _____
Date _____
Period _____

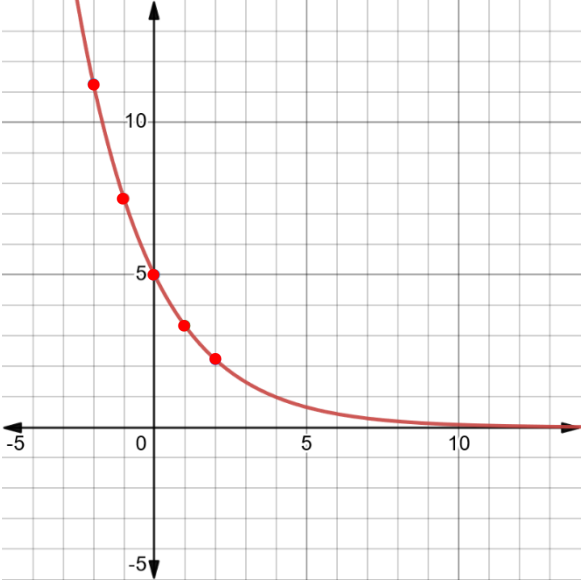
For questions 1 – 2, consider the function $f(x) = -4(1 + 0.65)^x$.

1. Since a is ☐ positive
☒ negative and b is ☐ between 0 and 1,
☒ greater than 1, the function
- is ☐ increasing.
☒ decreasing. The left end behavior of the graph is ☒ $y \rightarrow 0$
☐ $y \rightarrow -\infty$ and the
- right end behavior is ☐ $y \rightarrow 0$.
☒ $y \rightarrow -\infty$. The y -intercept of the function is ☒ -4 .
☐ 0.65 .
☐ 1.65 .

2. Sketch the graph of $f(x) = -4(1 + 0.65)^x$.



Graph the function of $y = 5\left(\frac{2}{3}\right)^x$, complete the table of values, and then determine all key features.

3. Graph	4. Table of Values												
	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>11.25</td></tr><tr><td>-1</td><td>7.5</td></tr><tr><td>0</td><td>5</td></tr><tr><td>1</td><td>$3\frac{1}{3}$ or $3.\bar{3}$</td></tr><tr><td>2</td><td>$2\frac{2}{9}$ or $2.\bar{2}$</td></tr></table>	x	y	-2	11.25	-1	7.5	0	5	1	$3\frac{1}{3}$ or $3.\bar{3}$	2	$2\frac{2}{9}$ or $2.\bar{2}$
x	y												
-2	11.25												
-1	7.5												
0	5												
1	$3\frac{1}{3}$ or $3.\bar{3}$												
2	$2\frac{2}{9}$ or $2.\bar{2}$												

Key Features	
5. Growth or Decay	decay
6. Domain and Range	$\mathbb{R}$ $y > 0$
7. Increasing/Decreasing	decreasing $(-\infty, \infty)$
8. Constant Percent Rate of Change	$\frac{2}{3}$ or $66\frac{2}{3}\%$
9. Left and Right End Behavior	as $x \rightarrow -\infty$, $y \rightarrow \infty$ as $x \rightarrow \infty$, $y \rightarrow 0$
10. Positive or Negative	positive only $(-\infty, \infty)$

Use the scenario below for questions 1 – 5.

Tom purchased a used Honda CRV from a dealership for \$15,575. The car depreciates 15% per year. The situation can be modeled by the function $P(x) = \$15,575(0.85)^x$ where x represents the number of years since Tom purchased the car and P is the value of the car.

1. Does the equation $P(x) = \$15,575(0.85)^x$ represent exponential growth or decay?

decay since $b < 1$

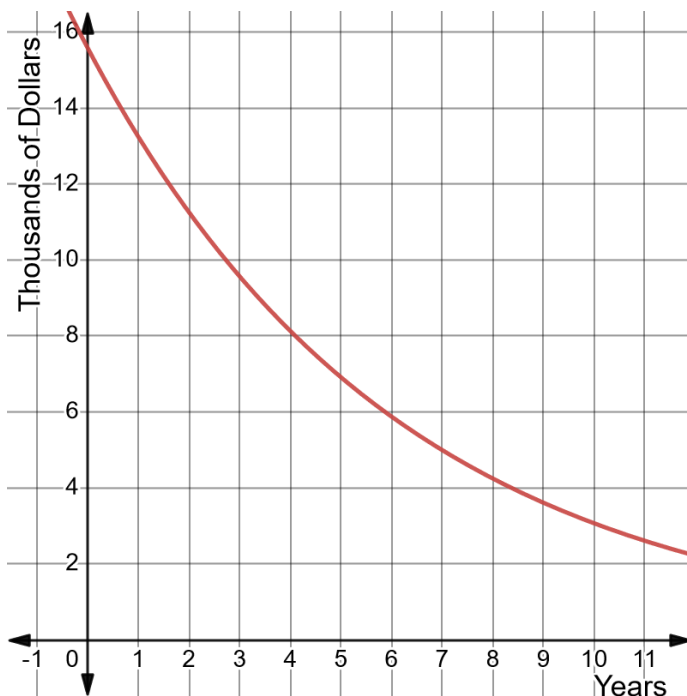
2. What is the y -intercept and what does it represent in this situation?

It represents the initial amount paid for the car.

3. Explain why the value of b is not 0.15, the rate of depreciation.

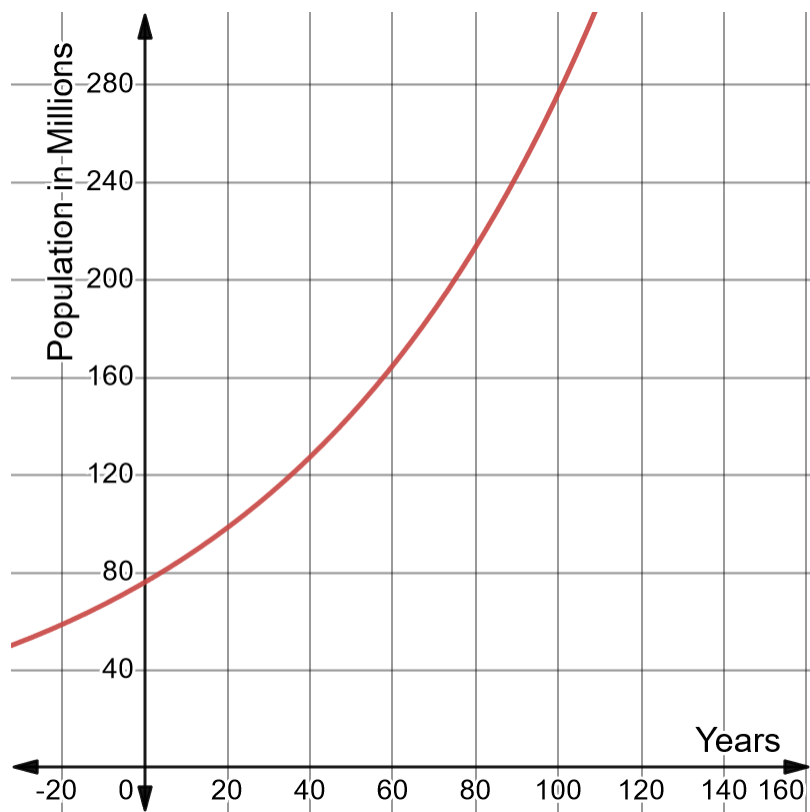
Because $1 - 0.15 = 0.85$ is what the car is still worth.

4. Graph the equation $P(x) = \$15,575(0.85)^x$.



The equation $f(t) = 76(1 + 0.013)^t$ represents the U.S. population and t represents the number of years after 1900. Graph the function below, then determine the key features and interpret how each relates to the context.

5.



Key Features		Interpretations
6. Domain	$x \geq 0$	The years after 1900
7. Range	$y \geq 76$	The population of the U.S.
8. Y-intercept	(0,76)	In 1900, the population was 76 million
9. Constant Percent Rate of Change	$1.013 = 101.3\%$	The population increased yearly by a rate of 101.3%.
10. End Behavior	As $x \rightarrow -\infty, y \rightarrow 0$ As $x \rightarrow \infty, y \rightarrow \infty$	The population can continue to grow exponentially over the years.

The graph of an exponential function is shown.

1. Find the value of a by substituting the point $(0, 2)$ in for x and y in $y = a(b)^x$.

$$[2] = a(b)^{[0]}$$

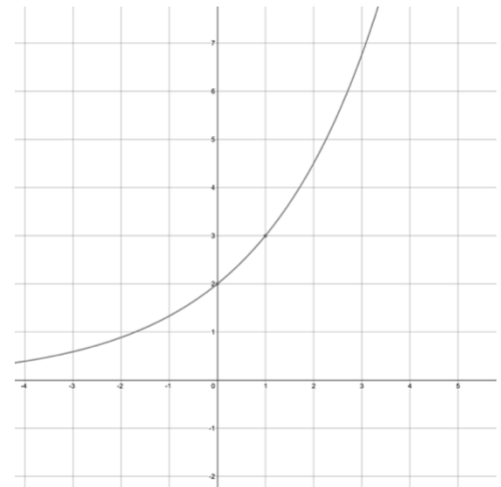
$$a = 2$$

2. Use the value of a found in question 1 and the point $(1, 3)$ to determine the value of b .

$$y = 2(b)^x$$

$$3 = 2(b)^1$$

$$b = 1.5$$



3. Write the function for the graph in both forms.

$f(x) = a(b)^x$	$f(x) = a(1 \pm r)^x$
$f(x) = 2(1.5)^x$	$f(x) = 2(1 + 0.5)^x$

The table of an exponential function is shown.

x	0	1	2	3	4
$f(x)$	-7	-14	-28	-56	-112

4. What is the initial value, a ?

$$-7 = a(b)^0$$

$$a = -7$$

5. How does the value of $f(x)$ change as the x -value increases by 1?

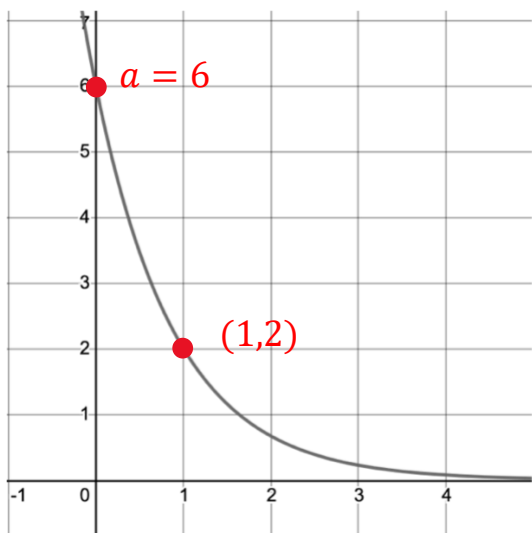
As the x -value increases by 1, the y -value or $f(x)$ value is multiplied by a factor of 2.

6. Write the function in the form $f(x) = a(b)^x$

$$f(x) = -7(2)^x$$

For questions 7 – 10, write an equation from the table or graph by finding the initial value, a , and common ratio, b .

7.



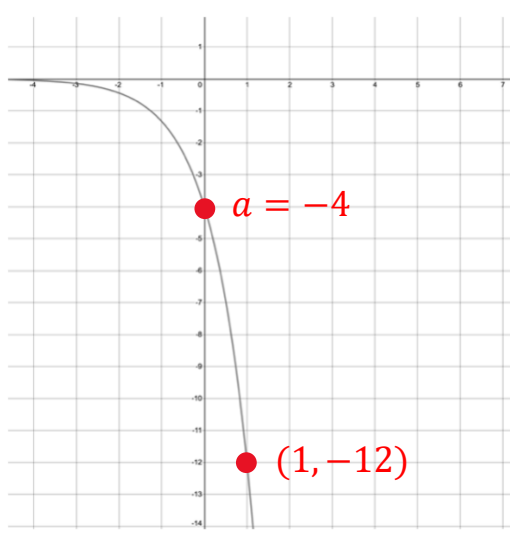
Function

$$2 = 6(b)^1$$

$$b = \frac{1}{3}$$

$$f(x) = 6\left(\frac{1}{3}\right)^x$$

8.



Function

$$-12 = -4(b)^1$$

$$b = 3$$

$$f(x) = -4(3)^x$$

9.

x	y
0	-5
1	$-\frac{25}{4}$
2	$-\frac{125}{16}$
3	$-\frac{625}{64}$
4	$-\frac{3125}{256}$

$a = -5$

$$\times \frac{5}{4}$$

$$\times \frac{5}{4}$$

$$\times \frac{5}{4}$$

$$\times \frac{5}{4}$$

Function

$$\frac{-\frac{25}{4}}{-5} = \frac{-5(b)^1}{-5}$$

$$b = \frac{5}{4} \text{ or } 1.25$$

$$f(x) = -5\left(\frac{5}{4}\right)^x \text{ or }$$

$$f(x) = -5(1.25)^x$$

10.

x	y
0	3
1	2.1
2	1.47
3	1.029
4	0.7203

$a = 3$

$$\times 0.7$$

$$\times 0.7$$

$$\times 0.7$$

$$\times 0.7$$

$$\times 0.7$$

Function

$$\frac{2.1}{3} = \frac{3(b)^1}{3}$$

$$b = 0.7$$

$$f(x) = 3(0.7)^x$$

Use the situation below to answer questions 1 – 3.

A new Microsoft Surface computer costs \$1275, but depreciates 16.5% each year as new technology comes out.

1. Is this situation exponential growth or decay?

Exponential decay

2. Two students, Josh and Eve were debating how to represent a function to model the value, V , after x years. Both of their functions are below. Explain who you agree with and why.

Josh	Eve
$V(x) = 1275(0.835)^x$	$V(x) = 1275(1.165)^x$

I agree with Josh because to calculate decay, subtract the rate from 1.
 $1 - 0.165 = 0.835$.

3. Write the correct function that models the context using both forms.

$V(x) = a(b)^x$	$V(x) = a(1 \pm r)^x$
$V(x) = 1,275(0.835)^x$	$V(x) = 1,275(1 - 0.165)^x$

Use the situation below to answer questions 4 – 6.

A new school opened with 225 students in 2021 and plans to increase by 13.3% per year until they reach full capacity.

4. Is this situation exponential growth or decay?

Exponential growth

5. Write an equation that models the population of the school, P , after x years since the opening of the school in 2021.

$P(x) = 225(1 + 0.133)^x$ or $P(x) = 225(1.133)^x$

6. Write the correct function that models the context using both forms.

$P(x) = a(b)^x$	$P(x) = a(1 \pm r)^x$
$P(x) = 225(1.133)^x$	$P(x) = 225(1 + 0.133)^x$

7. A three-bedroom house in the suburbs of Jacksonville sold for \$320,628. If the housing prices are expected to increase 2.8% annually, write a function that models the cost of a house, C , x years after it was sold.

$$C(x) = 320,628(1 + 0.028)^x \quad \text{or} \quad C(x) = 320,628(1.028)^x$$

8. A small local company reported a profit of \$175,000 last year but is struggling due to the pandemic. They are reporting a decrease of 13.6% each year. Write a function that represents the profit, P , after t years.

$$P(t) = 175,000(1 - 0.136)^t \quad \text{or} \quad P(t) = 175,000(0.864)^t$$

9. As of 2019, the world population is 8.215 billion people and is expected to grow at a rate of 1.2% per year. Write an equation to model the population growth, where $P(y)$ is the population in billions of people and y is the time in years.

$$P(y) = 8.215(1 + 0.012)^y \quad \text{or} \quad P(y) = 8.215(1.012)^y$$

10. A culture of 22 bacteria grows at a rate of 150% each hour. Tyler wrote the equation $B(t) = 22(1.5)^t$ to represent the population of the bacteria after t hours. Describe and correct the error.

Tyler should add 1 to 1.5.

$$B(t) = 22(1 + 1.5)^t \quad \text{or} \quad B(t) = 22(2.5)^t$$